Radial Flow: Diffusivity Equation Derivation

Notation Guide

Every derivative in this PDF is written out explicitly. The compact symbol ∇ (“del”) is not used.

Symbol	Meaning
$\partial p/\partial x$	how fast pressure changes with x, with the other variables held fixed (the slope of p in the x direction)
$\partial^2 p/\partial x^2$	the slope of the slope (curvature of p in x)
$\partial p/\partial t$	how fast pressure changes with time
[ρv] at x	the value of the product ρv evaluated at position x
Δx, Δt	small increments of position and time
η = k/(φμc_t)	hydraulic diffusivity
c, c_f, c_t	fluid, rock (formation) and total compressibility
p'	pressure drawdown (used in the gravity section)

Background: Building Blocks Used in the Derivation

Every diffusivity equation is built from the same three ingredients: a **mass balance** on a small control volume, **Darcy’s law** for the flow rate, and **compressibility** to link density and porosity to pressure. Only the shape of the control volume (and so the flow area) changes between linear, radial and spherical systems. This section sets out the shared parts once.

B1. Assumptions and variables

Assumptions:

1. Single-phase, slightly compressible fluid
2. Homogeneous, isotropic rock: k constant. Fluid viscosity μ constant
3. Darcy's law applies (laminar flow)
4. No gravity effects
5. Isothermal
6. Small pressure gradients

Variables:

- ρ = fluid density
- v = Darcy velocity (volume flow rate per unit area)
- φ = porosity
- p = pressure
- c = fluid compressibility
- c_f = rock (pore) compressibility. The f stands for **formation** (the rock), not fluid. With several fluids, fluid compressibility carries a phase subscript (c_o, c_w, c_g). With a single fluid it is written simply as c.

B2. Mass crossing a face

Mass = density × volume. The volume of fluid crossing a face of area A at Darcy velocity v during time Δt is A·v·Δt. So

$$\text{mass} = \rho\, A\, v\, \Delta t$$

Units check: (kg/m³)·m²·(m/s)·s = kg ✓

B3. Darcy's law (differential form)

Flow goes from high to low pressure, at a rate proportional to the local pressure gradient:

$$v = -\frac{k}{\mu}\frac{\partial p}{\partial s}$$

where s is the direction of flow. For flow toward the center (the −r direction), s = −r, so ∂/∂s = −∂/∂r and

$$v = +\frac{k}{\mu}\frac{\partial p}{\partial r}$$

Pressure increases with r, so fluid moves inward from high to low pressure ✓. This **point form** holds in any geometry and at any time. The familiar q = kAΔp/(μL) is its integrated form for constant area and steady flow.

B4. Compressibility and the accumulation term

The mass stored in a control volume ΔV is the pore volume times the density. Fluid fills only the pore space, so the pore volume is φΔV, not ΔV:

$$\text{mass stored} = \rho\ (\ \phi\, \Delta V)\ = \Delta V\ (\ \phi\rho\ )$$

The porosity φ does not appear in the mass in/out terms because the Darcy velocity is already defined per unit total (bulk) area.

Both ρ and φ depend on pressure. **Fluid compressibility** and **rock compressibility** are

$$c = \frac{1}{\rho}\frac{\partial \rho}{\partial p}\ \Rightarrow\ \frac{\partial \rho}{\partial p} = c\,\rho, \qquad c_f = \frac{1}{\phi}\frac{\partial \phi}{\partial p}\ \Rightarrow\ \frac{\partial \phi}{\partial p} = c_f\,\phi$$

Since ρ and φ depend on position and time only through p, the chain rule gives

$$\frac{\partial \rho}{\partial s} = c\,\rho\,\frac{\partial p}{\partial s}, \qquad \frac{\partial \rho}{\partial t} = c\,\rho\,\frac{\partial p}{\partial t}, \qquad \frac{\partial \phi}{\partial t} = c_f\,\phi\,\frac{\partial p}{\partial t}$$

Apply the product rule to the accumulation term and substitute:

$$\frac{\partial\ (\ \phi\rho\ )}{\partial t} = \phi\frac{\partial \rho}{\partial t} + \rho\frac{\partial \phi}{\partial t} = \phi\, c\rho\frac{\partial p}{\partial t} + \rho\, c_f\phi\frac{\partial p}{\partial t}$$

Factor out φρ ∂p/∂t and define the **total compressibility** c_t = c + c_f:

$$\frac{\partial\ (\ \phi\rho\ )}{\partial t} = \phi\,\rho\, c_t\,\frac{\partial p}{\partial t} \qquad (\ B4\ )$$

For multiphase flow, c_t = c_f + S_o c_o + S_w c_w + S_g c_g. For one liquid phase it reduces to c + c_f.

B5. Why a small term can be dropped (linearization)

Expanding the flow term always produces a nonlinear piece of the form c (∂p/∂s)², multiplied by c. For a slightly compressible liquid, c is of order 10^{-6} to 10^{-5} psi^{-1} and pressure gradients are moderate away from the wellbore, so this term is far smaller than the second-derivative term. Dropping it gives a **linear** equation that can be solved analytically. The accuracy of this step is lowest for gases, where c varies strongly with pressure (pseudo-pressure is used instead).

B6. Hydraulic diffusivity

$$\eta = \frac{k}{\phi\,\mu\,c_t}$$

In SI units η has units of m^2/s, the units of a diffusion coefficient. In oilfield units (k in md, μ in cp, c_t in psi^{-1}, t in hours):

$$\eta = 0.0002637 \frac{k}{\phi \mu c_t} \quad \left[\frac{\text{ft}^2}{\text{hr}}\right]$$

Radial Flow Diffusivity Equation

Convention: flow toward the well, i.e. in the **−r direction**, with mass balanced over a time interval **Δt**.

Radial Step 0: Geometry and assumptions

Consider a cylindrical reservoir of thickness h with a well at the center. Take a thin **annular shell** between radius r and r + Δr. The assumptions are those listed in the Background section, except that flow is purely radial.

The key point is that the flow area depends on r (in linear flow the cross-section w·h is constant). The area of a cylindrical surface at radius r is

$$A\,(\,r\,) \;= 2\pi r h$$

So the area at r + Δr is larger than at r. In linear flow the area w·h is constant, so it cancels. Here it cannot.

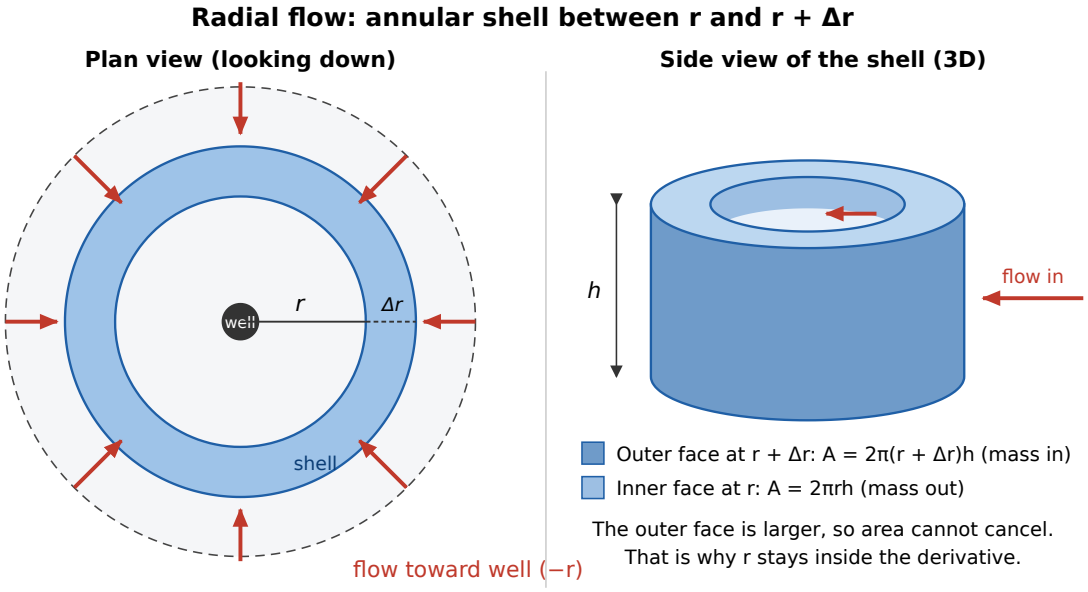


Figure 1. Radial flow. Left: plan view of the annular shell with flow converging on the well. Right: 3D view of the same shell. Fluid enters through the outer cylindrical face and leaves through the inner one, which has less area.

Radial Step 1: Mass crossing each face over Δt

Using mass = ρ A v Δt:

Mass in (outer face, at r + Δr):

$$(\text{ mass })_{in} = 2\pi h\, \Delta t\, [r\rho v]_{r+\Delta r}$$

Mass out (inner face, at r):

$$(\text{ mass })_{out} = 2\pi h\, \Delta t\, [r\rho v]_{r}$$

The radius r stays inside the bracket because it is part of the area, and the area changes between the two faces.

Radial Step 2: Mass accumulated in the shell

The bulk volume of the shell is the difference of two cylinders:

$$\Delta V = \pi h[\,(r+\Delta r)^2 - r^2] = \pi h(2r\,\Delta r + \Delta r^2)$$

When Δr is very small, Δr² is negligible compared with 2rΔr, so

$$\Delta V \approx 2\pi r h\, \Delta r$$

The accumulated mass is

$$(\text{ mass })_{acc} = 2\pi r h\, \Delta r[\,(\phi\rho)_{t+\Delta t} - (\phi\rho)_{t}]$$

Radial Step 3: Mass balance

(mass in) − (mass out) = (mass accumulated):

$$2\pi h\, \Delta t([\,r\rho v\,]_{r+\Delta r} - [\,r\rho v\,]_{r}) = 2\pi r h\, \Delta r[\,(\phi\rho)_{t+\Delta t} - (\phi\rho)_{t}]$$

Radial Step 4: Divide and take limits

Divide both sides by 2π h Δr Δt:

$$\frac{[\,r\rho v\,]_{r+\Delta r} - [\,r\rho v\,]_{r}}{\Delta r} = r\,\frac{(\phi\rho)_{t+\Delta t} - (\phi\rho)_{t}}{\Delta t}$$

Take Δr → 0 and Δt → 0:

$$\frac{\partial(r\rho v)}{\partial r} = r\,\frac{\partial(\phi\rho)}{\partial t}$$

Divide by r to get the **radial continuity equation**:

$$\frac{1}{r}\frac{\partial(r\rho v)}{\partial r} = \frac{\partial(\phi\rho)}{\partial t} \qquad (R1)$$

Compare with the linear case, ∂(ρv)/∂x = ∂(φρ)/∂t. The only change is the factor r inside the derivative, which comes from the growing area.

Radial Step 5: Darcy's law

For flow in the −r direction (Background B3):

$$v = +\frac{k}{\mu}\frac{\partial p}{\partial r} \qquad (R2)$$

Pressure increases with r, so fluid moves inward from high to low pressure ✓

Radial Step 6: Combine continuity and Darcy

Substitute (R2) into (R1) and pull the constants k and μ outside:

$$\frac{k}{\mu}\frac{1}{r}\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \frac{\partial(\phi\rho)}{\partial t} \qquad (R3)$$

Radial Step 7: Right-hand side (accumulation)

This is the result of Background B4, because it involves only time derivatives and compressibility:

$$\frac{\partial(\phi\rho)}{\partial t} = \phi\rho\, c_t \frac{\partial p}{\partial t} \qquad (R4)$$

where c_t = c + c_f.

Radial Step 8: Left-hand side (flow term)

We need the derivative of a product of **three** factors: r, ρ and ∂p/∂r. The product rule gives:

$$\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \rho\frac{\partial p}{\partial r} + r\frac{\partial p}{\partial r}\frac{\partial \rho}{\partial r} + r\rho\frac{\partial^2 p}{\partial r^2}$$

Use the chain rule with the compressibility definition, ∂ρ/∂r = c ρ ∂p/∂r:

$$= \rho\frac{\partial p}{\partial r} + r\, c\rho\left(\frac{\partial p}{\partial r}\right)^2 + r\rho\frac{\partial^2 p}{\partial r^2}$$

Now multiply by 1/r and factor out ρ:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \rho\left[\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + c\left(\frac{\partial p}{\partial r}\right)^2\right] \qquad (R5)$$

The term (1/r) ∂p/∂r is new. It exists only because of the radial geometry.

Radial Step 9: Combine both sides

Substitute (R5) and (R4) into (R3), cancel ρ, and multiply by μ/k:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + c\left(\frac{\partial p}{\partial r}\right)^2 = \frac{\phi \mu c_t}{k}\frac{\partial p}{\partial t} \qquad (R6)$$

This is nonlinear because of the squared gradient term.

Radial Step 10: Linearization

Drop c $(\partial p/\partial r)^2$, since c is small ($10^{-6}$ to 10^{-5} psi^{-1}) and the gradient is moderate:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} = \frac{\phi \mu c_t}{k}\frac{\partial p}{\partial t}$$

The left side is exactly (1/r) ∂/∂r (r ∂p/∂r). Expanding that product gives back the two terms. So the compact form is:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \frac{1}{\eta}\frac{\partial p}{\partial t}, \qquad \eta = \frac{k}{\phi \mu c_t} \qquad (R7)$$

Caution: the gradient is largest near the wellbore, so the neglected term is least accurate there at high drawdown. It is still the standard assumption behind most well-test analysis.

Oilfield units: with r in ft, t in hours, k in md, μ in cp and c_t in psi^{-1}:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \frac{\phi \mu c_t}{0.0002637\, k}\frac{\partial p}{\partial t}$$

Linear vs radial

	Linear	Radial
Flow area	constant (w·h)	2πrh, grows with r
Shell volume	w·h·Δx	about 2πrh·Δr
Continuity	∂(ρv)/∂x = ∂(φρ)/∂t	(1/r) ∂(rρv)/∂r = ∂(φρ)/∂t
Diffusivity equation	$\partial^2 p/\partial x^2 = (1/\eta)\, \partial p/\partial t$	$\partial^2 p/\partial r^2 + (1/r)\, \partial p/\partial r = (1/\eta)\, \partial p/\partial t$

If the flow were outward (injection), the sign of v in Darcy's law and the sign in the continuity equation both flip, and they cancel. The final equation is the same.

Notes: Common Questions and Nuances

N1. Where does R3 come from? (Differential vs integrated Darcy's law)

Darcy's law is the same in both geometries. The familiar form q = kAΔp/(μL) is the **integrated** form, valid only for constant area and steady flow. The derivation uses the **differential (point) form**, which holds in any geometry and at any time.

Differential form (flow in −r direction):

$$v = \frac{k}{\mu}\frac{\partial p}{\partial r} \qquad (R2)$$

Linear integrated form. Multiply by area, q = vA = (kA/μ) dp/dx. With constant A and steady q, integrate over length L:

$$q = \frac{kA\,\Delta p}{\mu L}$$

Radial integrated form. Here A = 2πrh, so q = (2πrhk/μ) dp/dr. Steady q is constant but r is not, so separate variables:

$$q\int_{r_w}^{r_e}\frac{dr}{r} = \frac{2\pi kh}{\mu}\int_{p_w}^{p_e} dp \quad \Rightarrow \quad q = \frac{2\pi kh\,(p_e - p_w)}{\mu\,\ln\,(r_e/r_w)}$$

A logarithm replaces the length L, but the differential form behind both is identical.

R3 itself is not Darcy's law alone. It comes from substituting (R2) into the radial continuity equation (R1). The geometry enters through R1 (the factor r from the area), not through Darcy:

$$r\rho v = r\rho\frac{k}{\mu}\frac{\partial p}{\partial r} \quad \Rightarrow \quad \frac{k}{\mu}\frac{1}{r}\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \frac{\partial(\phi\rho)}{\partial t}$$

The diffusivity equation describes **unsteady** flow, where q varies with position and time, so the steady-state integrated forms cannot be used. The integrated radial formula is the steady (or pseudo-steady) well inflow equation.

N2. The compact form in the radial linearization step

The claim is that the two expressions below are the same:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} \quad \text{and} \quad \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right)$$

Expand the compact form. Inside is a product of a = r and b = ∂p/∂r. The product rule, with ∂r/∂r = 1, gives

$$\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = r\frac{\partial^2 p}{\partial r^2} + \frac{\partial p}{\partial r}$$

Multiply by 1/r. In the first term r cancels with 1/r:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right) = \frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r}$$

Numerical check with p = r³. Expanded form: p'' + p'/r = 6r + 3r = 9r. Compact form: r p' = 3r³, d/dr gives 9r², times 1/r gives 9r. Both agree.

Why use the compact form:

1. r ∂p/∂r is proportional to flow rate (q = (2πkh/μ) r ∂p/∂r), so the equation says "the change of flow rate with r equals the storage change".
2. It keeps that same quantity, r ∂p/∂r, visible in the equation, which is the natural quantity to use in boundary conditions at the well.
3. It is the radial part of the second-derivative terms in the full cylindrical equation.

The extra term (1/r) ∂p/∂r is the geometry correction from the growing flow area. Far from the well, 1/r is small and the radial equation looks more and more like the linear one.

N3. Why r is inside the derivative (radial) but x is not (linear)

What sits inside the derivative is the **flow area**, not r or x as such.

- **Linear:** the area w·h is the same at both faces, so it factors out of the difference and divides away. Nothing x-dependent remains.
- **Radial:** the area is 2π(r+Δr)h at the outer face and 2πrh at the inner face. Only 2π and h are constants. The factor r is different at each face, so it must stay inside the difference. After dividing by Δr the difference becomes a derivative of the whole product rρv.

Leaving r outside would drop a term:

$$\frac{\partial\,(\,r\rho v\,)}{\partial r} = \rho v + r\frac{\partial\,(\,\rho v\,)}{\partial r} \;\neq\; r\frac{\partial\,(\,\rho v\,)}{\partial r}$$

The extra ρv is the physical effect of the area growing with r. Check: for steady flow with constant q, rρv is constant, so ∂(rρv)/∂r = 0, giving ρv = C/r (velocity rises toward the well). Then ∂(ρv)/∂r is not zero, so keeping r outside would be wrong.

General rule for a flow area A(s) along the path s:

$$\frac{1}{A}\frac{\partial\left(\ A\rho v\ \right)}{\partial s}=\frac{\partial\left(\ \phi\rho\ \right)}{\partial t}$$

Geometry	Area A(s)	Constant?	Result
Linear	w·h	yes, cancels	∂(ρv)/∂x
Radial	2πrh	no	(1/r) ∂(rρv)/∂r
Spherical	4πr²	no	(1/r²) ∂(r²ρv)/∂r

A tapered linear channel with area A(x) would keep A(x) inside the bracket for exactly the same reason.

2D and 3D Radial (Cylindrical) Diffusivity Equation

The 1D radial equation assumed that pressure depends on r only. In general, pressure can also vary around the well (angle θ) and along the well (vertical position z). The derivation is the same mass balance, now on a small block in **cylindrical coordinates** (r, θ, z). I derive the full 3D case first. The 2D cases then follow by removing one direction.

Velocities are taken positive in the +r, +θ, +z directions, so mass enters at the lower face of each pair and leaves at the upper one. The final result does not depend on this choice.

Cyl Step 0: The control volume

The element spans r to r + Δr, θ to θ + Δθ, and z to z + Δz.

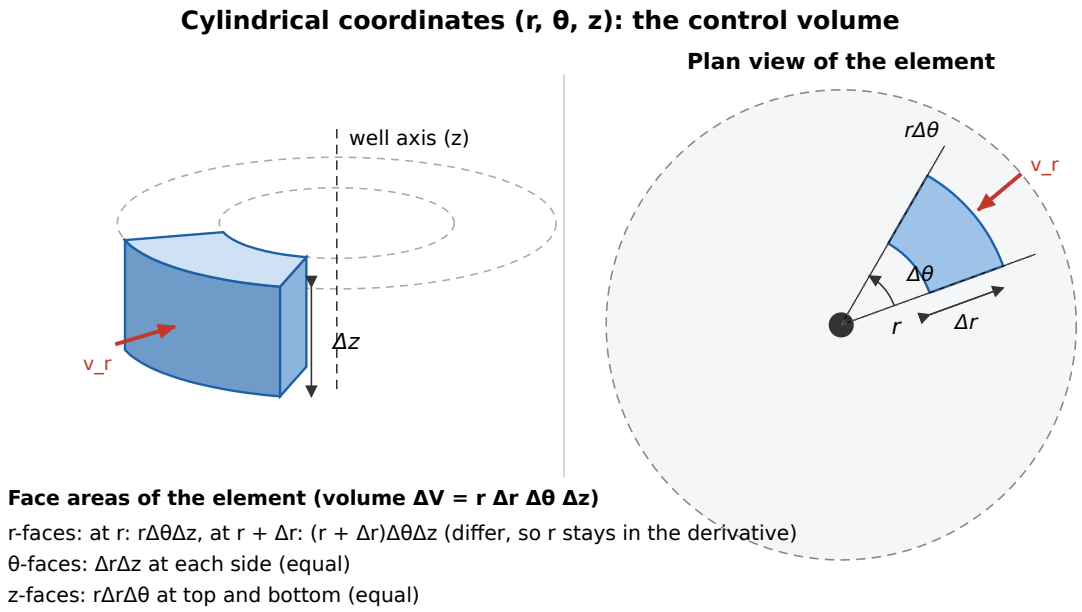


Figure 2. Cylindrical control volume. Left: 3D element. Right: plan view with the dimensions Δr, Δθ and the arc length rΔθ.

Its bulk volume is the area of the annular sector times the height. The sector has radial width Δr and arc length rΔθ (arc length = radius × angle in radians), so

$$\Delta V = r\,\Delta r\,\Delta\theta\,\Delta z$$

The six faces come in three pairs:

Pair	Face area	Same on both faces?
r-faces (normal to r)	at r: r Δθ Δz; at r + Δr: (r + Δr) Δθ Δz	no, grows with r
θ-faces (normal to θ)	Δr Δz	yes
z-faces (normal to z)	r Δr Δθ	yes, it depends only on r

The r-faces are the only pair whose area changes across the element. This is the same reason r stayed inside the derivative in the 1D radial case.

Cyl Step 1: Mass in and out over Δt

Using mass = ρ A v Δt on each face:

r-direction (in at r, out at r + Δr):

$$(\text{ mass })_{r,in} = \Delta\theta\,\Delta z\,\Delta t\,[\,r\rho v_r\,]_r, \qquad (\text{ mass })_{r,out} = \Delta\theta\,\Delta z\,\Delta t\,[\,r\rho v_r\,]_{r+\Delta r}$$

θ-direction (in at θ, out at θ + Δθ):

$$(\text{ mass })_{\theta,in} = \Delta r\,\Delta z\,\Delta t\,[\,\rho v_\theta\,]_\theta, \qquad (\text{ mass })_{\theta,out} = \Delta r\,\Delta z\,\Delta t\,[\,\rho v_\theta\,]_{\theta+\Delta\theta}$$

z-direction (in at z, out at z + Δz):

$$(\text{ mass })_{z,in} = r\,\Delta r\,\Delta\theta\,\Delta t\,[\,\rho v_z\,]_z, \qquad (\text{ mass })_{z,out} = r\,\Delta r\,\Delta\theta\,\Delta t\,[\,\rho v_z\,]_{z+\Delta z}$$

Here v_r, v_θ and v_z are the Darcy velocity components in the three directions.

Cyl Step 2: Accumulation

The mass stored is ρ times the pore volume φΔV, as before:

$$(\text{ mass })_{acc} = r\,\Delta r\,\Delta\theta\,\Delta z[\,(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_t]$$

Cyl Step 3: Mass balance

(total mass in) − (total mass out) = accumulation:

$$\begin{aligned}
&\Delta\theta\Delta z\Delta t(\,[\,r\rho v_r\,]_r - [\,r\rho v_r\,]_{r+\Delta r})\\
&+\Delta r\Delta z\Delta t(\,[\,\rho v_\theta\,]_\theta - [\,\rho v_\theta\,]_{\theta+\Delta\theta})\\
&+r\Delta r\Delta\theta\Delta t(\,[\,\rho v_z\,]_z - [\,\rho v_z\,]_{z+\Delta z})\\
&= r\Delta r\Delta\theta\Delta z[\,(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_t]
\end{aligned}$$

Cyl Step 4: Divide and take limits

Divide everything by Δr Δθ Δz Δt. In each term on the left, two of the three increments cancel and one stays in the denominator:

$$\begin{aligned}
&-\frac{[\,r\rho v_r\,]_{r+\Delta r} - [\,r\rho v_r\,]_r}{\Delta r} - \frac{[\,\rho v_\theta\,]_{\theta+\Delta\theta} - [\,\rho v_\theta\,]_\theta}{\Delta\theta}\\
&- r\,\frac{[\,\rho v_z\,]_{z+\Delta z} - [\,\rho v_z\,]_z}{\Delta z} = r\,\frac{(\,\phi\rho\,)_{t+\Delta t} - (\,\phi\rho\,)_t}{\Delta t}
\end{aligned}$$

Let all increments go to zero. Each difference quotient becomes a partial derivative:

$$-\frac{\partial\,(\,r\rho v_r\,)}{\partial r} - \frac{\partial\,(\,\rho v_\theta\,)}{\partial\theta} - r\,\frac{\partial\,(\,\rho v_z\,)}{\partial z} = r\,\frac{\partial\,(\,\phi\rho\,)}{\partial t}$$

Divide by r. Since r does not depend on θ or z, it passes through those derivatives:

$$-\frac{1}{r}\frac{\partial\,(\,r\rho v_r\,)}{\partial r} - \frac{1}{r}\frac{\partial\,(\,\rho v_\theta\,)}{\partial\theta} - \frac{\partial\,(\,\rho v_z\,)}{\partial z} = \frac{\partial\,(\,\phi\rho\,)}{\partial t} \qquad (\,C1\,)$$

This is the continuity equation in cylindrical coordinates. Compare with Cartesian: the only changes are the factor r inside the r-derivative and the 1/r on the θ term.

Cyl Step 5: Darcy's law in cylindrical coordinates

Darcy's law relates each velocity component to the pressure gradient **along that direction**. In r and z, the distances are dr and dz. In θ, the distance moved when the angle changes by dθ is the arc length **r dθ**, not dθ. So the gradient in the θ direction is (1/r) ∂p/∂θ:

$$v_r = -\frac{k}{\mu}\frac{\partial p}{\partial r}, \qquad v_\theta = -\frac{k}{\mu}\frac{1}{r}\frac{\partial p}{\partial \theta}, \qquad v_z = -\frac{k}{\mu}\frac{\partial p}{\partial z} \qquad (C2)$$

This is the source of the 1/r² in the final θ-term: one 1/r from Darcy and one from the continuity equation.

Cyl Step 6: Combine continuity and Darcy

Substitute (C2) into (C1). The three minus signs cancel, and k and μ come out of the derivatives. In the θ term, r is a constant with respect to θ, so it can be moved outside:

$$\frac{k}{\mu}\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \theta}\left(\rho\frac{\partial p}{\partial \theta}\right) + \frac{\partial}{\partial z}\left(\rho\frac{\partial p}{\partial z}\right)\right] = \frac{\partial(\phi\rho)}{\partial t} \qquad (C3)$$

Cyl Step 7: Expand each term

Radial term. Identical to Radial Step 8:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\rho\frac{\partial p}{\partial r}\right) = \rho\left[\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + c\left(\frac{\partial p}{\partial r}\right)^2\right]$$

θ term. Product rule on ρ ∂p/∂θ, then ∂ρ/∂θ = cρ ∂p/∂θ:

$$\frac{1}{r^2}\frac{\partial}{\partial \theta}\left(\rho\frac{\partial p}{\partial \theta}\right) = \frac{\rho}{r^2}\left[\frac{\partial^2 p}{\partial \theta^2} + c\left(\frac{\partial p}{\partial \theta}\right)^2\right]$$

z term. Same form as in 1D linear flow:

$$\frac{\partial}{\partial z}\left(\rho\frac{\partial p}{\partial z}\right) = \rho\left[\frac{\partial^2 p}{\partial z^2} + c\left(\frac{\partial p}{\partial z}\right)^2\right]$$

Accumulation term. Unchanged, because it involves only the time derivative:

$$\frac{\partial(\phi\rho)}{\partial t} = \phi\rho\, c_t \frac{\partial p}{\partial t}$$

Cyl Step 8: Combine, cancel ρ, and linearize

Substitute everything into (C3), cancel ρ, and multiply by μ/k:

$$\frac{\partial^2 p}{\partial r^2}+\frac{1}{r}\frac{\partial p}{\partial r}+\frac{1}{r^2}\frac{\partial^2 p}{\partial \theta^2}+\frac{\partial^2 p}{\partial z^2}+c\left[\left(\frac{\partial p}{\partial r}\right)^2+\frac{1}{r^2}\left(\frac{\partial p}{\partial \theta}\right)^2+\left(\frac{\partial p}{\partial z}\right)^2\right]=\frac{\phi\mu c_t}{k}\frac{\partial p}{\partial t}$$

The bracket multiplied by c contains the squared pressure slopes in the r, θ and z directions. Drop it for the same reason as before (c is small, gradients are moderate). The **3D radial (cylindrical) diffusivity equation** is:

$$\frac{\partial^2 p}{\partial r^2}+\frac{1}{r}\frac{\partial p}{\partial r}+\frac{1}{r^2}\frac{\partial^2 p}{\partial \theta^2}+\frac{\partial^2 p}{\partial z^2}=\frac{1}{\eta}\frac{\partial p}{\partial t},\qquad \eta=\frac{k}{\phi\mu c_t}$$

In compact form, the first two terms are (1/r) ∂/∂r (r ∂p/∂r), as shown in note N2:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial p}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 p}{\partial \theta^2}+\frac{\partial^2 p}{\partial z^2}=\frac{1}{\eta}\frac{\partial p}{\partial t}$$

2D radial cases

Removing one direction from the 3D equation gives two common 2D forms.

Case A: radial-vertical (r, z), axisymmetric flow

Pressure does not vary with angle, so all θ-derivatives vanish. The reservoir has thickness and flow has a vertical component. This applies to **partially penetrating wells, coning and vertical flow near the wellbore**:

$$\frac{\partial^2 p}{\partial r^2}+\frac{1}{r}\frac{\partial p}{\partial r}+\frac{\partial^2 p}{\partial z^2}=\frac{1}{\eta}\frac{\partial p}{\partial t}$$

With different horizontal and vertical permeabilities (k_r and k_z), Darcy's law uses the matching permeability in each direction, and the equation becomes:

$$\frac{\partial^2 p}{\partial r^2}+\frac{1}{r}\frac{\partial p}{\partial r}+\frac{k_z}{k_r}\frac{\partial^2 p}{\partial z^2}=\frac{\phi\mu c_t}{k_r}\frac{\partial p}{\partial t}$$

Case B: areal radial (r, θ), thin horizontal reservoir

Pressure does not vary with z (thin reservoir, no vertical flow), but may vary around the well. This applies to **non-uniform boundaries, interference from neighboring wells and angular pressure variation** in a horizontal layer:

$$\frac{\partial^2 p}{\partial r^2}+\frac{1}{r}\frac{\partial p}{\partial r}+\frac{1}{r^2}\frac{\partial^2 p}{\partial \theta^2}=\frac{1}{\eta}\frac{\partial p}{\partial t}$$

Recovering the 1D radial equation

If pressure depends on neither θ nor z, both extra terms vanish and the 1D radial equation (R7) is recovered:

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} = \frac{1}{\eta}\frac{\partial p}{\partial t}$$

Anisotropy in the 3D cylindrical case

With permeabilities k_r, k_θ and k_z, each Darcy component uses its own permeability:

$$k_r\left[\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r}\right] + \frac{k_\theta}{r^2}\frac{\partial^2 p}{\partial \theta^2} + k_z\frac{\partial^2 p}{\partial z^2} = \phi\mu c_t\frac{\partial p}{\partial t}$$

In most reservoirs the horizontal permeability is roughly uniform around a vertical well (k_θ = k_r), while k_z is smaller.

Gravity

When gravity is included, the vertical Darcy velocity has an extra term, v_z = −(k/μ)(∂p/∂z + ρg) with z measured upward. It was neglected above. For a slightly compressible liquid the gravity term is nearly constant in space and drops out at this level of approximation, so p can be read as pressure referred to a datum. The full treatment is in the section Including Gravity.

All coordinate systems side by side

System	Sum of second-derivative terms (the “Laplacian”)
1D Cartesian	$\partial^2 p/\partial x^2$
2D Cartesian	$\partial^2 p/\partial x^2 + \partial^2 p/\partial y^2$
3D Cartesian	$\partial^2 p/\partial x^2 + \partial^2 p/\partial y^2 + \partial^2 p/\partial z^2$
1D radial	$\partial^2 p/\partial r^2 + (1/r)\,\partial p/\partial r$
2D radial (r, z)	$\partial^2 p/\partial r^2 + (1/r)\,\partial p/\partial r + \partial^2 p/\partial z^2$
2D radial (r, θ)	$\partial^2 p/\partial r^2 + (1/r)\,\partial p/\partial r + (1/r^2)\,\partial^2 p/\partial \theta^2$
3D radial (r, θ, z)	$\partial^2 p/\partial r^2 + (1/r)\,\partial p/\partial r + (1/r^2)\,\partial^2 p/\partial \theta^2 + \partial^2 p/\partial z^2$

In every case the equation is (sum of second-derivative terms) = (1/η) ∂p/∂t. Only the geometry changes the form of the left side.

Including Gravity

The derivations above neglected gravity. Here it is included, writing every derivative out explicitly. The result is worth stating first:

- For a slightly compressible liquid, **gravity does not change the diffusivity equation**. It changes what the pressure variable means and what the initial condition is.
- The quantity that obeys the diffusivity equation is the **drawdown** p′ = p − p_i(z), where p_i(z) is the hydrostatic pressure before production. Its initial value is zero, and every solution for the no-gravity case applies to it.

The derivation below is written in Cartesian coordinates (x, y, z). Gravity acts only along the vertical direction z, so the steps that involve gravity are the same in any coordinate system. The sections after it show how this applies to the geometry of this PDF.

Gravity Step 1: Darcy's law with gravity

Let z be the **elevation, positive upward**, and g the gravitational acceleration. Without gravity the driving force was the pressure slope alone. With gravity, the weight of the fluid adds to the vertical direction only:

$$v_x = -\frac{k}{\mu}\frac{\partial p}{\partial x}, \qquad v_y = -\frac{k}{\mu}\frac{\partial p}{\partial y}, \qquad v_z = -\frac{k}{\mu}\left(\frac{\partial p}{\partial z} + \rho g\right)$$

Check: in static equilibrium, pressure increases downward at ∂p/∂z = −ρg, so v_z = 0 ✓.

Horizontal flow does not involve z, so gravity has no effect on it.

Gravity Step 2: Continuity equation (unchanged)

Gravity does not enter the mass balance, so the continuity equation is the same as in the 3D derivation. Net mass flowing out of a small block in the x, y and z directions equals the decrease of stored mass:

$$-\left[\frac{\partial\,(\,\rho v_x\,)}{\partial x} + \frac{\partial\left(\,\rho v_y\,\right)}{\partial y} + \frac{\partial\,(\,\rho v_z\,)}{\partial z}\right] = \frac{\partial\,(\,\phi\rho\,)}{\partial t}$$

Gravity Step 3: Substitute Darcy's law

The minus signs cancel, and k, μ and g are constants. The x and y terms look as before. The z term gains a gravity part, g ∂(ρ²)/∂z, because ρ times the extra term ρg gives ρ²g:

$$\frac{k}{\mu}\left[\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)+\frac{\partial}{\partial y}\left(\rho\frac{\partial p}{\partial y}\right)+\frac{\partial}{\partial z}\left(\rho\frac{\partial p}{\partial z}\right)+g\frac{\partial\left(\rho^2\right)}{\partial z}\right]=\frac{\partial\left(\phi\rho\right)}{\partial t}$$

Gravity Step 4: Expand each term

x, y and z pressure terms (product rule, with ∂ρ/∂x = cρ ∂p/∂x and so on). For x:

$$\frac{\partial}{\partial x}\left(\rho\frac{\partial p}{\partial x}\right)=\rho\frac{\partial^2 p}{\partial x^2}+c\rho\left(\frac{\partial p}{\partial x}\right)^2$$

The y and z terms have the same form.

Gravity part. Use ∂(ρ²)/∂z = 2ρ ∂ρ/∂z and ∂ρ/∂z = cρ ∂p/∂z:

$$g\frac{\partial\left(\rho^2\right)}{\partial z}=2c\rho^2 g\frac{\partial p}{\partial z}$$

Accumulation part (unchanged): ∂(φρ)/∂t = φρ c_t ∂p/∂t.

Gravity Step 5: Combine, cancel ρ, multiply by μ/k

$$\frac{\partial^2 p}{\partial x^2}+\frac{\partial^2 p}{\partial y^2}+\frac{\partial^2 p}{\partial z^2}+c\left[\left(\frac{\partial p}{\partial x}\right)^2+\left(\frac{\partial p}{\partial y}\right)^2+\left(\frac{\partial p}{\partial z}\right)^2\right]+2c\rho g\frac{\partial p}{\partial z}=\frac{1}{\eta}\frac{\partial p}{\partial t}\qquad(G1)$$

Compared with the no-gravity case, there is one extra term, 2cρg ∂p/∂z.

Gravity Step 6: The hydrostatic initial state

Before production there is no flow, so v_z = 0 and ∂p/∂z = −ρg. To first order (ρ constant over the height of the reservoir):

$$p_i\left(z\right)=p_{ref}-\rho g\ \left(z-z_{ref}\right)$$

Pressure is therefore **not uniform** initially. It varies with elevation. Define the drawdown as the departure from this state:

$$p'\left(x,y,z,t\right)=p-p_i\left(z\right),\qquad p'\left(x,y,z,0\right)=0$$

Gravity Step 7: Substitute the drawdown into (G1)

The hydrostatic part depends on z only, so the x and y slopes are unchanged, and ∂p_i/∂z = −ρg gives

$$\frac{\partial p}{\partial x}=\frac{\partial p'}{\partial x},\qquad\frac{\partial p}{\partial y}=\frac{\partial p'}{\partial y},\qquad\frac{\partial p}{\partial z}=\frac{\partial p'}{\partial z}-\rho g$$

For the second derivatives and the squared slope in z:

$$\frac{\partial^2 p}{\partial z^2} = \frac{\partial^2 p'}{\partial z^2} + \frac{d^2 p_i}{dz^2}, \qquad \left(\frac{\partial p}{\partial z}\right)^2 = \left(\frac{\partial p'}{\partial z}\right)^2 - 2\rho g \frac{\partial p'}{\partial z} + (\rho g)^2$$

Also ∂p/∂t = ∂p′/∂t, because p_i does not depend on time. The x and y terms of (G1) are unchanged apart from p becoming p′. The z terms of (G1) become:

$$\frac{\partial^2 p'}{\partial z^2} + \frac{d^2 p_i}{dz^2} + c\left(\frac{\partial p'}{\partial z}\right)^2 - 2c\rho g \frac{\partial p'}{\partial z} + c(\rho g)^2 + 2c\rho g \frac{\partial p'}{\partial z} - 2c(\rho g)^2$$

Two cancellations happen:

1. **The cross terms** −2cρg ∂p′/∂z and +2cρg ∂p′/∂z cancel each other.
2. **The static terms.** Differentiating the hydrostatic condition dp_i/dz = −ρg once more gives d²p_i/dz² = −g dρ/dz = c(ρg)². So d²p_i/dz² + c(ρg)² − 2c(ρg)² = 0.

What remains is exactly the no-gravity equation, written for the drawdown:

$$\frac{\partial^2 p'}{\partial x^2} + \frac{\partial^2 p'}{\partial y^2} + \frac{\partial^2 p'}{\partial z^2} + c\left[\left(\frac{\partial p'}{\partial x}\right)^2 + \left(\frac{\partial p'}{\partial y}\right)^2 + \left(\frac{\partial p'}{\partial z}\right)^2\right] = \frac{1}{\eta}\frac{\partial p'}{\partial t}$$

(The density in the gravity term is the local density, which differs from the hydrostatic density by a relative amount of order c·p′, about 10^{-4}. The cancellation is accurate to that order.)

Now linearize as before, by dropping the small bracket multiplied by c:

$$\frac{\partial^2 p'}{\partial x^2} + \frac{\partial^2 p'}{\partial y^2} + \frac{\partial^2 p'}{\partial z^2} = \frac{1}{\eta}\frac{\partial p'}{\partial t}, \qquad p'(x, y, z, 0) = 0$$

Gravity Step 8: Equivalent view with flow potential

Define the flow potential Φ = p + ρ g z. For constant ρ, Darcy's law becomes

$$v_x = -\frac{k}{\mu}\frac{\partial \Phi}{\partial x}, \qquad v_y = -\frac{k}{\mu}\frac{\partial \Phi}{\partial y}, \qquad v_z = -\frac{k}{\mu}\frac{\partial \Phi}{\partial z}$$

At equilibrium Φ is uniform:

$$\Phi_i = p_{ref} + \rho g\, z_{ref}$$

and Φ − Φ_i equals p′. So Φ obeys the diffusivity equation with a **uniform** initial value, which is why potential is often used for problems with significant vertical flow.

In practice: solve for p′ with the formulas from the no-gravity derivation, then add back the hydrostatic part to get the pressure at a given elevation: p = p_i(z) + p′.

Gravity Step 9: An independent size check

Even without the cancellation above, the gravity term is small. Take water with ρg ≈ 0.433 psi/ft and c ≈ 3 × 10⁻⁶ psi⁻¹:

$$2c\ (\ \rho g\ )^{\ 2} \approx 2\ \left(\ 3 \times 10^{-6}\ \right)\ \ (\ 0.433\ )^{\ 2} \approx 1.1 \times 10^{-6}\ \mathrm{psi/ft}^{2}$$

A typical diffusion term for a 100 psi drawdown over 100 ft is ∂²p/∂z² ≈ 100/100² = 0.01 psi/ft². The gravity term is about 10⁻⁴ of that, the same negligible order as the c(slope)² terms already dropped.

Gravity in Radial Flow

Horizontal radial flow (r, θ). The reservoir is a horizontal layer, so ∂z/∂r = 0 and ∂z/∂θ = 0. There is no gravity term in the radial derivation or in the areal (r, θ) equation.

Vertical plane (r, z). Gravity appears only in the vertical Darcy velocity, v_z = −(k/μ)(∂p/∂z + ρg). In cylindrical coordinates the full equation (G1) reads

$$\frac{\partial^2 p}{\partial r^2} + \frac{1}{r}\frac{\partial p}{\partial r} + \frac{\partial^2 p}{\partial z^2} + c\left[\left(\frac{\partial p}{\partial r}\right)^2 + \left(\frac{\partial p}{\partial z}\right)^2\right] + 2c\rho g\frac{\partial p}{\partial z} = \frac{1}{\eta}\frac{\partial p}{\partial t}$$

The hydrostatic pressure p_i(z) depends on z only, so substituting p = p_i(z) + p′ produces exactly the same cross-term and static-term cancellations as in Step 7. The result is

$$\frac{\partial^2 p'}{\partial r^2} + \frac{1}{r}\frac{\partial p'}{\partial r} + \frac{\partial^2 p'}{\partial z^2} = \frac{1}{\eta}\frac{\partial p'}{\partial t}$$

This is the 2D (r, z) equation from the cylindrical section, now written for the drawdown. The same holds for the 3D (r, θ, z) form, because θ is a horizontal direction and carries no gravity term.

Where gravity does matter. In coning and gas-cap problems the vertical forces on two fluids of different density compete with the pressure gradient. That needs a multiphase treatment, with a separate Darcy's law for each phase, and is outside the single-phase equation derived here.

When gravity cannot be dropped

1. **Gas.** Density varies strongly with pressure, so the hydrostatic gradient is not constant and the term 2cρg ∂p/∂z is not small. Pseudo-pressure with a gas-gradient correction is used.
2. **Multiphase flow.** Oil, gas and water have different densities, so gravity segregation drives vertical flow (coning, gas-cap expansion, gravity drainage).

Each phase has its own Darcy's law with its own ρg.

3. **Strongly dipping or very thick reservoirs.** The potential difference across the pay zone can matter in water-drive and gravity-drainage problems.
4. **Datum conversion.** When you compare wells at different depths, convert all pressures to a common datum using the fluid gradient. In field units the gradient is $0.433 \cdot \gamma$ psi/ft, where γ is the fluid specific gravity.